

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Reasons why state provided pension costs may be rising:**

There could be demographic impacts, e.g.

- increasing longevity ...
- which is worsened if combined with a falling birth rate, and
- birth booms in the past.

[½ mark for each]

The effect is increased if individuals have a shorter working lifetime due to:

- a lower proportion of the working population to the retired population
- more time spent in higher education
- career breaks
- earlier retirement.

[1 mark for each]

The extent of the impact on costs will depend upon whether the scheme is financed by PAYG or a funded approach. [½]

The impact on costs will be more noticeable if the scheme is financed by a PAYG method. [½]

Also need to consider whether contributions are earnings-linked or not. [½]

If the scheme is funded, then the rise in costs may reflect poor investment returns earned on the fund. [½]

Costs may increase if inflation-linked benefits are offered and inflation has been higher than expected. [½]

The lack of private provision may be due to:

- concerns about the security and risks of such provision [½]
- a lack of awareness about the importance of making provision [½]
- a lack of provision leading to a lack of competition in the market and high costs,
- further reducing private provision [½]
- belief that the State will provide sufficient benefits [½]
- low wages, so no spare money to make savings [½]
- a lack of tax and other incentives (relative to other savings vehicles) [½]
- a lack of incentives for employers or other providers to offer schemes [½]
- lack of paternalism by employers, eg. if the workforce is highly mobile [½]
- over-regulation of occupational schemes, making employers less inclined to offer them, due to the time and costs associated with compliance [½]
- over-regulation of private providers, eg. solvency margins etc [½]
- high charges (for personal pensions or occupational schemes). [½]

[Max 10]

ii) Advantages of the proposal

- if combined with reducing State benefits it can control State costs
- general inertia means many people may remain in the scheme even though an opt-out option is offered, thus increasing provision and overall savings
- the opt-out option is more politically acceptable than if the arrangement is compulsory
- easily copes with contribution holidays, career breaks, change of employment

- requiring employer contributions should help ensure a reasonable amount of money is saved, leading to meaningful benefits
- requiring only modest employer contributions should not be too onerous a burden on employers
- State operated and therefore, expected to be secure
- a funded arrangement so may have positive impact on investment markets, creates demand for investments
- offering investment choice enables individuals to tailor investments to their risk appetite and circumstances
- Defined Contribution in nature and therefore transparent and easy to understand
- funded and therefore likely to be more secure
- there should be benefits from economies of scale, ie lower administration and commission costs

Disadvantages of the proposal

- does not cover long-term unemployed, long-term carers, sickness
- not compulsory, and so does not ensure employed individuals have private provision
- unpopular as it may “appear” to be compulsory
- may have a negative impact on existing private provision, eg will existing final salary schemes be more likely to close?
- does not guarantee the pension since Defined Contribution in nature and the investment risk and longevity risk falls on the individual ...
- ... eg there is a risk of poor investment returns over the period to retirement or a fall in value at retirement
- ... eg there is a risk of the conversion terms at retirement being worse than expected due to longevity improvements
- individuals may feel they do not have sufficient knowledge to make investment choice, or may just make inappropriate choices....
- ... leading to the need for a default option or individuals paying for investment advice
- contributions made may not be sufficient to provide an adequate pension
- may be unpopular with employers as required to contribute
- there may be a need to offer some guarantee or underpin to the pension, with associated costs
- if the State offers tax incentives as encouragement then there is an associated cost
- a significant change to the system with associated costs, eg education will be required and a new administration system
- additional regulation will be needed
- there may not be a large enough market to support the buyout of benefits at retirement
- may not lead to a higher level of savings.

[½ mark per point, Max 10]

iii) Areas where regulations may be imposed:

- Operational – there may be restrictions on how such arrangements are set up, ie under trust or contract. There may be more protections if the scheme is set up under trust, eg assets held separately from the sponsor. [1]
- Maximum contribution level (sponsor and/or employee) – there may be restrictions on the level of contributions, in particular if tax incentives are given. The maximum level of contributions may be age-related to reflect the cost of accrual increasing with age (provided investment return exceeds earnings inflation). [1]
- Minimum contribution level (sponsor and/or employee) – a minimum level of contributions could be prescribed to try to ensure adequate benefit provision. [½]

- Taxation of contributions, investments, benefits – regulations may be imposed relating to the tax treatment of contributions, investment income and capital gains, pensions and any lump sum cash payment. [1]
- Minimum level of benefits – may be required to target a minimum level of benefits, or schemes could be required to provide some form of defined benefit underpin. [½]
- Type of investments – restrictions on investment choice could be imposed, eg limits on self-investment or the use of derivatives. [1]
- Form of benefit provision – the form of benefit provision could be prescribed, eg require the pension to be index-linked. Regulations may allow some amount of the benefit to be commuted for cash. [1]
- Age at which benefits may be taken – minimum and maximum ages of retirement may be prescribed. [½]
- Provision of partner benefits – may require provision of a partner's pension if the member has a partner. [½]
- Marketing of private benefit provision – regulation could be imposed on the quality of promotional material and the form of any illustrations. [½]
- Level of charges – legislation could impose a maximum level of charges by the pension fund manager and pension administrator. [½]
- Disclosure of information – legislation may outline the frequency and nature of disclosure of information. [½]
- Advisors – requirement to take advice or for the advisor to be properly qualified and/or regulated. Possibly a maximum commission agreement. [½]
- Regulation of finances of providers – eg solvency margins for fund managers and insurance companies. [½]

[½ mark per point, **Max 10**]

[30 Marks]

Solution 2:

- i) Disadvantages of unfunded schemes:
- No funds are set aside, so benefits are less secure
 - No flexibility in the pace of funding the benefits
 - The costs are not very stable
 - Tax benefits may not be available

Moving to funded approach

- Sudden large injection of cash
- Need to set up a separate trust, with additional regulatory requirements
- Additional governance and compliance requirements, leading to additional costs
- Additional burden of investments
- Tax benefits may not be available for initial contributions for past service

[½ mark for each point, **Max 4**]

ii) Rationale

- a) Provision of information about the scheme's operations helps identify poorly run schemes, e.g. for schemes with poor funding level or inappropriate or very risky investment strategy, members can then apply pressure through unions or other forums to press for change or make alternative / additional provision

- b) These restrictions limit exposure to more risky investments,
And avoid losses that the scheme may not be able to bear
Minimum holding in government bonds is because these are a secure type of asset
- c) Central insolvency should provide members with additional security
Given this source of finance is independent of the scheme or employer
- d) Requirement should ensure that funds are adequate
If not, action can be taken to address shortfall
So, members can have confidence that there are sufficient assets set aside

[½ mark for each point, **Max 4**]

iii)

a) Disclosure requirements

Advantages:

- Operation of scheme may be more accountable
- Members will be better informed about the scheme and so will be more confident
- The sponsor and trustees have freedom to set investment and funding strategies, subject only to appropriate disclosures
- Independent advisor may be able to review the scheme based on the disclosures

Disadvantages:

- Even with the disclosures, these regulations alone do not require the situation to be improved
- May be difficult to understand the information that are provided
- The information disclosed may be inadequate
- Additional costs associated with complying with these regulations

b) Investment restrictions

Advantages:

- Limits exposure to more risky investments that may jeopardize members' security e.g. volatile investments or self-investment
- Government bonds may be a close match to some of the scheme's liabilities

Disadvantages:

- Holding 40% in government bonds is unlikely to be appropriate for all schemes
- Likely to reduce expected returns
- This will ultimately increase employer's contributions
- Reduced diversification of scheme assets
- May have impact on investment and bond markets

c) Central solvency fund

Advantages:

- Highly secure for the benefits, as covered by insurance
- Economies of scale can be used to the advantage of members

Disadvantages:

- Insurance may not cover all events
- May encourage less prudent funding and investment policy
- Can lead to cross subsidy from better run schemes
- Costs likely to spiral during recession when there are likely to be more claims

d) **Minimum funding requirement**

Advantages:

- Increases overall security for members
- Helps build a robust pensions market

Disadvantages:

- Money can be put to better use
- Can lead to additional compliance burden for smaller schemes

[½ mark for each point, **Max 12]****[20 Marks]****Solution 3:**

i)

- Basic requirement is that there is a reasonable volume of stable, consistent data, from which future experience and trends can be deduced. Since the scheme is medium sized, the data seems to be reasonable, however, given that there has been Covid deaths, hence they may need to be excluded as one-off event [1]
- The data ideally needs to be divided into sufficiently homogenous risk groups, according to the relevant risk factors (e.g. age, sex). [½]
- However, this has to be balanced against the danger of creating data cells that have too little data in them to be credible. [½]
- For example, for benefit schemes with a small number of members, it may not be appropriate to carry out any analysis or at least the results should be recognised as being very crude. [½]
- This may also be true when events are infrequent and volatile which is the case here. [½]
- For example, a benefit scheme with only young members may have many members but few deaths in service and so an analysis of the proportion of deaths may lack credibility. Here it is not mentioned how the scheme is distributed among the active and pensioners. [½]
- In practice the level of detail in the classification of the data depends upon the volumes of data available. The volume of data will not only indicate whether or not an analysis will produce meaningful results, but it will also indicate the extent to which data can be subdivided without leading to similar problems. [½]
- For example, it may be necessary to group data on deaths into, say, five-year age bands rather than single year bands. [½]
- As well as data on the feature being assessed, it is necessary to have data on the exposed to risk, divided into the same cell structure as the experience data. An analysis of experience is not valid unless experience and exposed to risk are matched. [½]
- Once the data have been grouped in an appropriate manner, the analysis can be performed. For statistical factors, such as mortality and withdrawal, this will involve the calculation for each age band of the number of deaths (or withdrawals) divided by the number exposed to risk of death (or withdrawal). [½]
- The results can then be compared with the assumptions adopted to determine whether there is a significant difference and also with other relevant standard tables to determine if they appear to be more appropriate to the scheme. [½]

[6]

ii)

- The results of an analysis of experience done for the first scheme should not be used blindly
- The major difference here in the analysis is the size itself which for a smaller scheme may not be appropriate to use
- Unlikely that the groups would be credible
- There are only 50 members in the scheme so will be very difficult to put into suitable groups.
- May be so few deaths (or even none) in a given period that you cannot treat the results as reliable.
- If the experience indicates that mortality is heavy, this could be influenced by one or two deaths – any change in assumption based on this would not be advisable.
- These deaths could have occurred for a number of reasons including covid deaths so may need to explore whether relevant to make changes to assumptions.
- If all of the 50 lives are of the same age then this may be useful to compare against standard tables and then adjust for some of the experience.

[6]

iii)

- May be better not to carry out the exercise and use standard tables instead. [½]
- Adjust standard table to reflect known characteristics of membership e.g. by pincode or region. [½]
- Could use other schemes mortality information and group the experience such that the results become more credible. [½]
- But occupations of the other schemes may be different so need to consider if relevant compared to standard tables. [½]

[2]

iv)

- The results of any analysis should not be used blindly [½]
- Consideration should be given to whether the period under investigation was typical [½]
- For example there may have been a change in legislation or abnormal event in the investigation period which encourages / discourages take up of the option [½]
- Or a change in the factors to make them more/less generous [½]
- Or the results may be skewed by the option of one member with a significant pension [½]
- Or there may be significant random fluctuations or errors in data [½]
- There may be a gradual change in experience over the period [½]
- And care should be taken to identify whether past trends are likely to continue in the future [½]
- Consideration should be given to the quality of the data [½]
- For example whether there is sufficient data for the analysis to be credible [½]
- A large scheme is likely to have enough data to perform credible analysis [½]
- But this may not be the case at all ages [½]
- Whether the data is recent enough to be relevant [½]
- Consideration should be given to whether experience is homogeneous across all groups or whether further grouping of data is appropriate [½]
- For example if the 15% taking no cash are correlated [½]
- e.g. in time / pension size [½]
- The actuary may recommend reducing the assumption to reflect experience [½]
- The basis used to set commutation factors is often less prudent than the funding basis so that if members do take cash, there is an experience gain. [½]
- Therefore the current assumption is not prudent based on experience [½]

- There may be additional elements of prudence elsewhere in the basis [½]
- And the materiality of changing the assumption may mean any changes are not significant... [½]
- for example if there is little difference in the assumptions used for funding and setting commutation terms [½]

[Max 6]

v) Advantages

- Scheme managers are concerned with the security of benefits and so could prevent members taking commutation if this would jeopardise the security of the fund [½]
- For example if the value of the cash taken is higher than the value of the benefits surrendered [½]
- Such as if commutation terms are fixed and the market related valuation basis would be weaker than the basis used to set the factors [½]
- Or to prevent members selecting against the scheme e.g. if they are in ill health [½]
- May aid in liquidity planning [½]
- Scheme managers are independent of company so would not be influenced by company concerns in approving/disapproving commutation requests [½]

Disadvantages

- May reduce take-up rate of the option [½]
- Which may reduce security of remaining benefits as risk remains in the scheme rather than being reduced [½]
- This is likely to cause an increase costs. For example: [½]
- Will add administrative complexities [½]
- And may require legal advice / amendment of the scheme rules [½]
- It will also occupy significant trustee time given the size of the scheme [½]
- Role is also to ensure fairness between all members [½]
- Therefore it would not be appropriate to treat individuals differently on a case by case basis [½]
- There is precedent in the scheme for members taking cash [½]
- So it may be difficult to withhold consent in practice [½]
- And may lead to complaints / disgruntled members [½]

[Max 5]

[25 Marks]

Solution 4:

i)

- Liabilities of future spending (and any debt), and size of planned inheritance. [½]
- What is the required standard of living. [½]
- Need to consider nature - fixed vs real - and term of liabilities - eg planned retirement date, dependants [½]
- What currency are they in? For example, what if family are educated overseas or individual plans to retire abroad. [1]
- What tax position is he in? Differing tax regimes may apply during income and inheritance phases. [½]
- How much liquidity is needed? [½]
- Potential for changing circumstances/liabilities into the future. [½]
- Investor's risk appetite. [½]
- Information should be obtained by an appropriate method (eg interview client, study financial reports). [½]

[½ for each point, Max 4]

- ii) Since he is covered by protection products in case of untimely death, the main risks/concern is that their fund is exhausted earlier than expected. This could be because of:
- Longevity (of individual and dependents) - living longer than expected [½]
 - Taxation, risk of changes, or ability to exploit rules made to encourage individuals to save - i.e. higher than expected taxes [½]
 - Lower than expected return on assets or volatility leads to insufficient income in retirement/liquidity problems or if inheritance too low. [½]
 - Higher than expected increase of liabilities relative to assets. [½]
 - Expenses of investment [½]
- [½ for each point, **Max 2**]

- iii)
- Longevity - invest part of portfolio in protection product such as annuity (with spouse's pension). [1]
 - Monitor health, take advantage of products offering protection e.g. term product which, as per the details in the question, he has already had. [1½]
 - Taxation - take advice. Diversify between asset classes which may have different tax treatment. [1]
 - E.g. maximise allowances for both income and capital gains, or tax exempt products. [1]
 - Asset returns - diversify by asset class, currency, use an investment manager [½]
 - Relative growth of liabilities - Matching, including de-risk the strategy over time as investment time horizon shortens, regular monitoring and take action. [1]
 - Expenses - consider collective investment schemes compared with direct investment; choose passive funds over active if appropriate [½]
 - As long as funds have low tracking errors [½]
 - Always look at net returns (active may be worth paying for in certain cases) [½]
- [Max 6]**
[12 Marks]

Solution 5:

- i)
- The valuation method for the assets should be consistent to that used for the liabilities e.g. if the market value is taken for the assets then the liabilities should be calculated using a consistent rate. Here the assets were confirmed to be valued at Market value but there is no information about the liabilities valuation.
 - The nature of the liabilities, heterogeneous benefits and scheme membership means that no "market value" is available i.e. there is not a liquid secondary market for trading existing pension scheme liabilities.
 - The purpose of the valuation is important. If it is to calculate the funding level e.g. for accounting or regulatory purposes or for a commercial transactions e.g. bulk transfers using market value is appropriate.
 - However, the market value is volatile in the short term which is an issue in setting a stable contribution level and may cause credibility problems. This said deficit might have happened due to the volatility of the market value and subsequent decision of paying NIL contribution in the fund.
 - If the scheme is a going concern the definition of market value used i.e. willing seller and willing buyer, should be consistent with the fact that the assets will not be immediately liquidated.
 - A misleading picture may be created, for example if the funding basis uses market values and funding levels are then quoted.
 - The main advantage is that the market value is readily available and also easily understood and communicated e.g. trustees. However there may be practical problems if no or multiple market values are available e.g. property or unquoted shares, no market value versus other assets which have more than one market value.

[4]

ii)

- The Actuary might have suggested the DCF approach as the discounted cash flow allows consistency with the ongoing e.g. cash transaction liability calculation but requires more calculation and is hard to estimate future cashflows.
- Neither may be an alternative if they are not permitted by local regulations. e.g., DCF is subjective in nature e.g., choice of assumptions and discount rates vary from one Actuary to the other.

[3]

iii) **General points:**

- The whole economy is in recession.
- There was a boom i.e. returns were very high for a period, rather than being particularly bad now.
- There is currently a weak market and market values are low. This has been seen across all sectors.
- The economic environment may have changed so all asset classes have lower absolute returns.
- The growth in capital values has reduced and income streams are lower.
- Due to the changing demographics there has been more demand for long term investments such as bonds. The type and/or term of investment has changed, for example less equity demand due to lack of stability.
- Poor political climate, uncertainty, riots, unemployment etc.
- Unfavourable economic factors such as rising interest rates
- Abundance of other investments, for example safer government bonds
- Unstable inflation
- New regulation due to banking crises so shift in investments
- Tax and regulation with suitable examples e.g. removal of dividend tax credit?

Scheme Specific

- A comparison of absolute returns achieved between different periods is not a good guide for how the investment manager has performed relative to their benchmark or peers. The investment managers may have had brief periods of underperformed in the prior 10 years, but out performed in the rest of the period,
- The fund manager may have achieved higher returns by taking higher risk in the equity portfolio in the prior 10 years. Or the fund manager may have changed. Due to low funding levels the fund requires less volatile investments. To reduce the volatility the level of risk and therefore expected returns within the equity portfolio may have been reduced, for example lower beta equities, and sectors subject to less volatile returns, no unquoted equities.
- The investment strategy has been changed for example from an active to a passive one i.e. tracking the market rather than taking risk for higher returns.
- There have been regulatory changes. These have changed the way in which equities are valued. The new approach to valuing equities has reduced the returns compared to the previous valuation method (assumptions).
- These changes have required that assets be disposed of and different ones purchased. This in itself incurs expenses but also may have been done at inopportune times. Possible crystallisation of capital gains tax due to changes in investment strategy.

[6]

[13 Marks]
